

50: Enhancement of a Surgical Robotics Public Engagement Demo - using da Vinci Surgical Tools and a 7DOF Serial Arm Manipulator

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Project Keywords (up to 5 keywords):

Robotics, ROS2, Mechatronics, System Integration, da Vinci Surgical Tools

Background:

Surgical robotics is a key component in modern healthcare, enabling minimally invasive procedures with improved precision, clinician ergonomics, and patient outcomes. Systems such as the Intuitive Surgical da Vinci surgical platform are widely used in clinical practice, yet the principles behind these technologies remain largely inaccessible to the general public. Public engagement plays a vital role in medical research by increasing transparency, fostering trust in technologies, and inspiring future engineers and clinicians. Interactive robotic demonstrations provide an effective means of communicating complex surgical robotics concepts in an intuitive and engaging manner.

The Robotics and Vision in Medicine (RViM) lab has previously developed a portable public engagement demonstrator combining: a custom control interface for da Vinci EndoWrist surgical instruments; a lightweight 7-degree-of-freedom serial robotic arm (Ufactory xArm 7); teleoperation using a gaming controller and ROS2 framework for combined control.

While functional, the current setup has limitations in usability, robustness, and clarity for non-expert audiences. This project aims to enhance the demonstrator's technical reliability, interaction design, and educational value, enabling its use in open days, school outreach, public exhibitions, and stakeholder visits.

Objectives and Methods:

The objective of this project is to enhance the existing surgical robotics public engagement demonstrator by improving its robustness, usability, and overall interaction quality for non-expert users. The work will primarily focus on refining the robotic control pipeline, including velocity control, kinematic behaviour, and the underlying ROS 2 software architecture and workflow, but may also include physical design improvements.

The student will analyse the current system to identify limitations in responsiveness, stability, and ease of use, and will implement software-level improvements to teleoperation, motion scaling, safety mechanisms, and also system documentation. Depending on progress and student interest, the project may also include minor physical or mechanical refinements to the setup to improve reliability and presentation for repeated public use. The enhanced system will be validated through testing in demo scenarios, with a focus on smooth and reliable operation, safety, and demonstration quality.

Skills and Requirements:

This project will involve hands-on work with a surgical robotics demonstrator, including robotic teleoperation, system integration, and software development using ROS 2, Arduino, and associated robotic control software. The student will work with a 7-DOF robotic arm, da Vinci surgical instruments, and associated control software, gaining experience in robotic kinematics, velocity control, and safety in human-robot interaction. The project will also develop skills in communicating complex robotic systems through well-designed demonstrations.

No prior experience in surgical robotics is strictly required. However, students should have a basic understanding of robotic control and kinematics, as well as programming skills (e.g. Python and/or C++) and an interest in robotics or mechatronic systems. It is beneficial to have a familiarity with ROS2/ROS.

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